

3. The image below shows sweet peas which can be a number of different colours including white and purple.



Pollen was transferred from the anthers of white flowers onto the stigmas of purple flowers. In the F₁, some plants produced purple flowers and some produced white flowers.

Two hypotheses have been suggested to explain this result.

- (a) The **first hypothesis** is that the purple variety is caused by a dominant allele of a single gene.
- (i) Construct a genetic diagram to show the genotypes and phenotypes of the parents of this cross. Use the letters **A** and **a** to represent the alleles of this gene. [2]



- (ii) Based on the first hypothesis, the F1 would be expected to contain equal numbers of plants producing purple flowers compared to white flowers. When the cross was carried out, the resulting seeds were planted and 32 plants produced white flowers and 18 produced purple flowers.

State the null hypothesis and complete the table below. [4]

Null hypothesis

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Phenotype	Observed numbers (O)	Expected numbers (E)	O-E	(O-E) ²	$\frac{(O-E)^2}{E}$
White	32				
Purple	18				

Use the formula below to calculate the value of chi-squared (χ^2).

$$\chi^2 = \sum \frac{(O-E)^2}{E}$$

$\chi^2 =$

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(iii) Use the table of chi-squared values below to state whether you would accept or reject the null hypothesis stated in (a)(ii). Explain your answer. [4]

Degrees of freedom	Probability								
	0.9	0.8	0.7	0.5	0.2	0.1	0.05	0.02	0.01
1	0.016	0.064	0.15	0.46	1.64	2.71	3.84	5.41	6.64
2	0.21	0.45	0.71	1.39	3.22	4.60	5.99	7.82	9.21
3	0.58	1.00	1.42	2.37	4.64	6.25	7.82	9.84	11.34
4	1.06	1.65	2.20	3.36	5.99	7.78	9.49	11.67	13.28

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- (b) The **second hypothesis** is that the purple variety is produced by an interaction between two unlinked genes **A** and **B**. The presence of at least one dominant allele for both genes results in purple flowers. If the plant is homozygous recessive for either gene, the phenotype will be white.
- (i) Complete the Punnett square to determine the phenotype ratios expected on the basis of this second hypothesis, given that the genotype of the purple flower is **AaBb** and the genotype of the white flower is **aaBb**. The gametes for the white flowering plant have been done for you. [3]

aB				
ab				

Phenotype ratio

- (ii) The value of chi-squared for the second hypothesis was calculated as 0.044. Using the data conclude which hypothesis is more likely to be correct, the first or the second. Explain your answer. [1]
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- (c) Explain how a cross between two white flowered parents which produced purple flowered offspring would confirm that hypothesis 2 is correct. [1]
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